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23ADL401 DEEP LEARNING • CASE STUDY: PAPER IMPLEMENTATION

Classifying Brain Tumors on MRI by Using Convolutional Neural Networks

Marco Antonio Gómez-Guzmán et al. • *Electronics, 2023*

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Presentation Outline

01 The Paper
Details, problem, motivation

02 Proposed Method
Architecture and key equations

03 Dataset & Setup
Data, tools, hyperparameters

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Pipeline and challenges faced

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Critique, learnings, references

Paper at a Glance

Item	Details
Title	Classifying Brain Tumors on Magnetic Resonance Imaging by Using CNNs
Authors	Gómez-Guzmán, Jiménez-Beristain, García-Guerrero, López-Bonilla, Tamayo-Perez, Esqueda-Elizondo, Palomino-Vizcaino, and Inzunza-González
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Paper first page



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Article

Classifying Brain Tumors on Magnetic Resonance Imaging by Using Convolutional Neural Networks

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Abstract: The study of neuroimaging is a very important tool in the diagnosis of central nervous system tumors. This paper presents the evaluation of seven deep convolutional neural networks (CNN) models for the task of brain tumor classification. A generic CNN model is implemented and compared with the CNN models created by the transfer learning (TL) technique. The dataset is Nickparvar's Brain Tumor MRI Dataset, which includes Glioblastoma, Sarcoma, Granuloma, and Normal tumors, containing 7023 MRI images. The magnetic resonance imaging (MRI) is the clinical technique to four classes, three brain tumors, and one healthy. Glioblastoma, Sarcoma, and Granuloma are the tumors. The seven CNN models included in this study were MRI Inception, ResNet50, MobileNetV2, InceptionResNetV2, InceptionV3, NASNetMobile, and EfficientNetB0. In the comparison of all CNN models, including a generic CNN and six pre-trained models, the best CNN model for this dataset was InceptionV3, which obtained an accuracy of 99.24%. The results of the implementation of these techniques could help clinicians specializing in the early detection of brain tumors.

Keywords: brain tumor; neuroimaging; image classification; deep learning; transfer learning; CNN; MRI; artificial intelligence

1. Introduction

A brain tumor (also abbreviated as BT) is an abnormal growth of brain cells that may be cancerous or noncancerous [1]. The brain is the most complex organ in the body and is made of normal cells [2]. Benign BTs are composed uniformly of inactive cells. In contrast, malignant tumors are composed of cancerous, active cells with a non-uniform structure [3]. Malignant BTs can be primary or secondary tumors. Malignant BTs can be primary tumors, the cancerous cells are contained within the brain; however, in the case of metastatic tumors, the cancerous cells have spread to other parts of the body, not just the brain [4,5]. Because it affects people of all ages, BT is of the deadliest illnesses in the world [6]. It is the most prevalent malignancy in older persons and the third most prevalent in children and teenagers [7]. Gliomas are the most common BTs, and they originate in the Pituitary [8]. Gliomas are found in certain brain organs, such as the cerebral pedicle and spinal cord. They induce symptoms such as vomiting, headaches, and discomfort. They are categorized as Low-Grade Gliomas (LGG) and High-Grade Gliomas (HGG) tumors [9]. Glioma cases are growing at an alarming rate with a serious impact on human mortality. Similarly, a Meningioma is a tumor that forms in the meninges, layers that surround the brain and spread cord. A Pituitary tumor is an irregular enlargement that forms in the Pituitary gland. In most cases, these tumors are benign [10].

Problem & Motivation

Problem Addressed

Classify MRI images into glioma, meningioma, pituitary tumor, or no-tumor. The study tests whether CNNs can perform this four-class image classification reliably.

- Supports analysis of large MRI collections.
- Early classification may assist clinical assessment; this study is not clinical validation.

Limitations of Earlier Work

- Several prior studies used only two or three classes.
- Datasets and evaluation protocols varied across studies.

Key contribution: compare one generic CNN and six pretrained models on a combined, four-class dataset of 7,023 MRI images.

Background Concepts

CNN feature extraction (Module 3)

Four convolution blocks learn hierarchical image features; ReLU adds non-linearity and max pooling reduces spatial dimensions.

Data augmentation (Module 2)

The reported pipeline uses rotation, horizontal flip, shear, zoom, and brightness transformations to vary training images.

Transfer learning (Module 3)

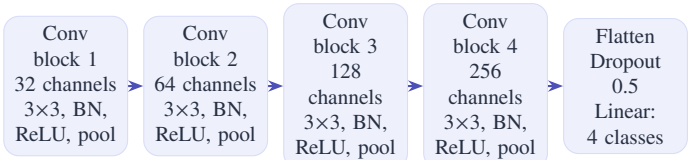
The paper compares ImageNet-pretrained backbones with a generic CNN trained from scratch; pretrained classifiers are adapted to four output classes.

Softmax and cross-entropy (Module 1)

Softmax maps four logits to class probabilities; cross-entropy compares them with the target label during training.

Module mapping: 23ADL401, 2023 Autonomous Scheme.

Proposed Method: Architecture



E2

PaperGenericCNN implementation; input $256 \times 256 \times 3$, 651,044 parameters. The paper does not specify every filter count; the schedule shown is the implemented schedule.

How it works

- RGB MRI input: 256×256
- Four blocks: 32, 64, 128, 256 channels
- BatchNorm, ReLU, and 2×2 max pooling
- Flatten, dropout, and four-class linear head

Paper compares the generic CNN with six pretrained CNNs.

Key Equations & Loss Function

No.	Operation	Equation	Meaning
1	Core operation	$Y = W * X + b$; $\mathrm{ReLU}(z) = \max(0, z)$	Convolution extracts local features; ReLU adds nonlinearity.
2	Loss function	$L = -\sum_{c=1}^{4} y_c \log p_c$	E2 uses PyTorch CrossEntropyLoss with no label smoothing.
3	Output / update	$p_c = e^{z_c} / \sum_{j=1}^{4} e^{z_j}$	E2 uses Adam at 10^{-4}; paper settings for the generic CNN conflict.

Note: The paper does not specify a loss implementation for every model; the stated PyTorch loss is the E2 setting.

Dataset

Item	Paper	Our implementation
Dataset name	Msoud: Figshare, SARTAJ, Br35H	Masoud Nickparvar Brain Tumor MRI (Kaggle)
No. of samples	7,023 original; 9,139 after augmentation	7,200 raw: 5,600 Training + 1,600 Test
No. of classes	4: Glioma, Meningioma, No-tumor, Pituitary	Same four classes
Train / Val / Test	80/20, 70/30, and 5-fold CV are reported	4,480/1,120 deterministic stratified split from Training; 1,486 clean Testing images quarantined
Input size	Generic CNN: 256×256; pretrained sizes vary	$256 \times 256 \times 3$ (E2 Generic CNN)
Preprocessing	1/255, rotation, flip, shear, zoom, brightness $[0, 0.7]$	1/255; rotation 10°, flip $p = 0.5$, shear 10°, zoom $[0.9, 1.1]$, brightness $[0, 0.7]$

Class counts in paper

Class	Train	Test
Glioma	1,321	300
Meningioma	1,339	306
No-tumor	1,595	405
Pituitary	1,457	300

Our balanced split: 1,120 training and 280 validation images per class.

Leakage audit: 111 exact-duplicate groups; 114 test duplicates excluded. The 1,486 clean test images remain quarantined (0 loaded, 0 evaluated).

Implementation Setup

Paper setup

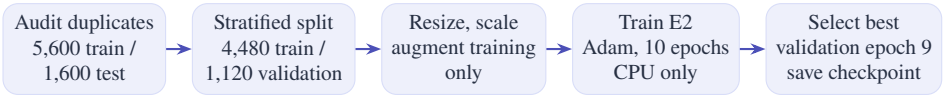
- Python 3.8.10
- TensorFlow 2.9.2
- Google Colab
- NVIDIA A100-SXM4-40GB

Our setup (E2)

- Python 3.12.7; PyTorch 2.14.1+cpu
- Arch Linux x86_64, kernel 7.1.5
- Intel Core i7-1255U (10c/12t); CPU only
- 7.4 GiB RAM; modular PyTorch codebase

Hyperparameter	Paper	Ours (E2)
Optimizer	Transfer: SGD; generic: Adam in Table 2 / SGD in text	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	10^{-4}	0.0001 (10^{-4})
Batch size	16	16
Epochs	10 per fold	10; stratified validation split
Dropout	Transfer: 0.3–0.4; generic: 0.5 in table / 0.2 in text	0.5 (Figure 2 setting)
Loss / regularization	Paper settings vary / incompletely specified	CrossEntropyLoss; no label smoothing or weight decay

Implementation Pipeline



Implementation record

PaperGenericCNN: 651,044 parameters; input $256 \times 256 \times 3$; four convolution blocks; dropout 0.5; four-class linear head.

Training: 4,480 samples; validation: 1,120 samples; best checkpoint selected at epoch 9.
Test: 1,486 clean images, quarantined and not evaluated.
Code: GitHub repository; training script; configuration.

Challenges Faced & Solutions

Paper and implementation challenges

Challenge	How it was solved
Paper gives conflicting generic-CNN optimizer and dropout values.	E2 follows Adam (10^{-4}) from Table 2 and dropout 0.5 from Figure 2; the paper's conflict remains a reproducibility limitation.
Duplicate leakage across dataset splits.	Pixel-hash audit identified 111 exact-duplicate groups; 114 test duplicates were excluded; clean test data stayed quarantined.
CPU-only compute and limited memory.	PyTorch ran 10 epochs in about 61.4 minutes on an i7-1255U, CPU only; report states about 606 MB peak footprint.

Brightness augmentation: possible contributor

The paper-literal brightness range $[0, 0.7]$ can strongly darken images. It may contribute to the Meningioma false-positive pattern, but was not isolated experimentally. E3 (safe brightness jitter $[0.7, 1.3]$) is proposed future work, not a completed experiment.

Results: Paper vs Our Implementation

Reported paper results and E2 validation results

Metric	Paper Generic CNN	E2 Generic CNN (Val)	Δ vs Generic	Paper InceptionV3
Accuracy	81.08%	80.98%	−0.10 pp	97.12%
Precision	85.27%	85.33%	+0.06 pp	97.97%
Recall	76.77%	80.98%	+4.21 pp	96.59%
Macro-F1	Not reported	81.86%	Not available	Not reported
Cross-entropy loss	0.4661	0.8291	+0.3630	0.0796
Parameters	8.7M	651,044	−8.049M	23.9M
Training time	200 min	61.4 min	−138.6 min	323 min

Like-for-like baseline

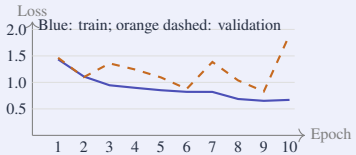
E2 accuracy is 0.10 percentage points below the paper's Generic CNN. Precision and recall differ by +0.06 and +4.21 points, respectively. This is close baseline performance, not proof of identical implementation.

Evaluation scope

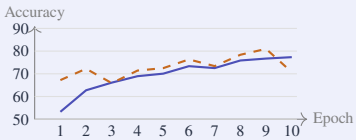
Paper values are reported model results; E2 values are from the best validation epoch ($N = 1, 120$). The 1,486 clean test images were not evaluated. InceptionV3 is a separate pretrained model, not a like-for-like baseline.

Training Curves & Sample Outputs

Loss by epoch



Accuracy by epoch (%)



Epoch 9 confusion matrix — validation

Actual \ Pred.	Glioma	Mening.	No tumor	Pituitary	Total
Glioma	218	58	0	4	280
Meningioma	14	247	5	14	280
No tumor	3	64	211	2	280
Pituitary	1	48	0	231	280
Predicted total	236	417	216	251	1,120

Epoch 10 confusion matrix — validation

Actual \ Pred.	Glioma	Mening.	No tumor	Pituitary	Total
Glioma	104	172	3	1	280
Meningioma	0	271	6	3	280
No tumor	0	38	242	0	280
Pituitary	0	103	2	175	280
Predicted total	104	584	253	179	1,120

E2 Epoch History

Complete 10-epoch history — Configuration E2

Epoch	Train loss	Train acc.	Train macro-F1	Val loss	Val acc.	Val macro-F1	Duration
1	1.4350	53.26%	0.5341	1.4650	67.23%	0.6742	6m 17s
2	1.1114	62.75%	0.6280	1.0994	72.23%	0.7127	6m 11s
3	0.9460	66.12%	0.6613	1.3590	65.89%	0.6562	6m 06s
4	0.8974	68.95%	0.6899	1.2429	71.43%	0.7019	6m 07s
5	0.8524	70.07%	0.7007	1.0941	72.50%	0.7084	6m 07s
6	0.8216	73.37%	0.7343	0.8767	76.34%	0.7559	6m 06s
7	0.8206	72.52%	0.7251	1.3858	73.39%	0.7269	6m 03s
8	0.6866	75.87%	0.7589	1.0374	78.39%	0.7878	6m 05s
9	**0.6520**	76.72%	0.7674	0.8291	**80.98%**	**0.8186**	6m 09s
10	0.6701	77.34%	0.7733	1.8991	70.71%	0.7099	6m 11s

E2: Adam, 10^{-4} learning rate, batch 16, dropout 0.5; 61.4 minutes total. Epoch 9 is the validation-selected checkpoint (80.98% accuracy, 81.86% macro-F1). Test images were not evaluated.

Analysis & Discussion

Why results differ

- E2 is a 651k-parameter CNN trained from scratch; InceptionV3 has 23.9M parameters and ImageNet pretraining.
- Against the paper's Generic CNN, E2 accuracy differs by only -0.10 percentage points.
- Paper reports model results from its protocol; E2 uses a stratified validation split. The test set remained quarantined.

Per-class metrics ($n = 280$ each)

Class	Prec.	Recall	F1
Glioma	92.37%	77.86%	84.50%
Meningioma	59.23%	88.21%	70.88%
No tumor	97.69%	75.36%	85.08%
Pituitary	92.03%	82.50%	87.01%
Macro avg.	85.33%	80.98%	81.86%

Meningioma errors

Epoch 9 predicted 58 gliomas, 64 no-tumor, and 48 pituitary cases as meningioma. Reported filenames are Training-source examples, not verified validation predictions.

Sample files

Training-source examples: Glioma `Tr-gl_101.jpg`; Meningioma `Tr-aug-me_2.jpg`; no-tumor `Tr-no_1007.jpg`; Pituitary `Tr-pi_1006.jpg`. Reported error examples: Glioma `Tr-gl_10.jpg`; no-tumor `Tr-no_1020.jpg`; Pituitary `Tr-pi_1115.jpg`. Validation membership and predictions are unverified.

Epoch 10 / E3

Epoch 10: validation accuracy 70.71% (loss 1.8991); 584/1,120 predictions were Meningioma. Cause is unknown. Brightness dimming is a hypothesis; E3 jitter $[0.7, 1.3]$ remains untested.

Critical Review of the Paper

Strengths

- Compares seven CNN models.
- Includes four classes, including no-tumor.
- Reports multiple metrics and model size.

Limitations

- Dataset availability and computation are constraints.
- Split and generic-CNN settings are ambiguous.
- No external clinical validation is reported.

Future Scope

- Make data split and augmentation protocol explicit.
- Report class-wise results.
- Test on an independent dataset.

Conclusion & Key Learnings

Conclusion

- The paper compared seven CNNs; InceptionV3 reported 97.12% accuracy.
- E2 achieved 80.98% validation accuracy and 81.86% macro-F1 ($N = 1,120$).
- E2 is within 0.10 percentage points of the paper's Generic CNN accuracy; this does not establish identical training or evaluation.
- The clean 1,486-image test set remains unevaluated; no clinical validation is claimed.

What I Learned

- Medical-image augmentation choices can affect diagnostic features and should be tested through controlled ablations.
- Pixel-hash audits and strict test quarantine help prevent data leakage.
- CPU-only training can reproduce a compact baseline, while evaluation protocol and pretrained architecture must be considered in comparisons.

References & Resources

References

[1] M. A. Gómez-Guzmán et al., “Classifying Brain Tumors on Magnetic Resonance Imaging by Using Convolutional Neural Networks,” *Electronics*, vol. 12, art. 955, 2023. DOI: 10.3390/electronics12040955.

[2] Masoud Nickparvar, “Brain Tumor MRI Dataset,” Kaggle.

[3] Figshare brain tumor dataset, SARTAJ brain tumor MRI dataset, and Br35H, as cited by Gómez-Guzmán et al.

Implementation artifacts

Repository: github.com/qwerty123950/neuroscan_ai.
Training script: train.py; config: train_config.yaml.
Referenced checkpoint and metrics: `models/checkpoints/paper_generic_cnn_best.pt`; `reports/metrics/paper_generic_cnn_training_summary.json`. Artifacts are not present in this workspace.

Thank You

Questions & Discussion